

A FROZEN ULW BACKBONE

PSG epoch sequence



t-2 t-1 t t+1 t+2

- EEG
- EOG
- EMG

ULW-SleepNet
channel-shared
DSSC
frozen

Local evidence

$$h_t \in \mathbb{R}^{128}$$

$$z_t^{\text{ULW}} \in \mathbb{R}^5$$

B LINEAR-BIM (+3.1K params)

Projection

$$128 \rightarrow 16$$

$$p_t = \tanh(W_p h_t)$$

Past context

3 learned scales

$$\tau_1 \bullet \tau_2 \bullet \tau_3 \bullet$$

$$m^- \quad d^- = p - m^-$$

Future context

3 learned scales

$$\tau_1 \bullet \tau_2 \bullet \tau_3 \bullet$$

$$m^+ \quad d^+ = p - m^+$$

Temporal vector

$$q_t \in \mathbb{R}^{208}$$

$$[p \mid m^+ \mid d^+]$$

Linear correction

$$\Delta z_t = W q_t + b$$

$$z^{\text{ULW}} + \Delta z_t$$

$$\rightarrow \hat{y}_t$$

C BIM-EXPLAIN · EXACT

Residual → 8 exact groups

$$\Delta z_t = L_t + \sum_{k=1}^3 (H_{k,t}^- + H_{k,t}^+) + b$$

■ local

■ past₀₋₂

■ future₀₋₂

■ bias

exact to numerical precision

Mechanism evidence

① **Margin** $M_{\text{final}} = M_{\text{ULW}} + \sum_g \Delta M_g$

② **True history = 0** → $\hat{y}$ flips

③ **Innovation** GT transition ↔ stable

corrected · harmed · N1 · subject bootstrap